

Speech Intelligibility Project

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Section 1: Read in Data and Load Packages

First, we load the necessary packages for reading in the data and then load the dataset. We expect there to be 115 children.

Load necessary packages

```
# --- Data I/O & Wrangling ---
library(readxl)      # read Excel files
library(tidyverse)   # data wrangling and visualization (dplyr, ggplot2, tidyr, purrr)

# --- Reporting & Tables ---
library(knitr)        # render output tables and format reports
library(table1)       # formatted descriptive (Table 1) summaries

# --- Statistical Modeling ---
library(tidymodels)   # model utilities (broom for tidy coefficient output)
library(lmerTest)     # linear mixed models with Satterthwaite p-values
library(emmeans)      # estimated marginal means and pairwise contrasts
library(broom.mixed)  # tidy() method for lme4/lmerTest mixed-effects model objects

# --- Statistical Testing & Diagnostics ---
library(irr)          # inter-rater reliability (ICC)
library(ggpubr)       # normality diagnostic plots (density, Q-Q)
library(TOSTER)       # equivalence testing (TOST, bootstrapped)
```

Read in Data

```
speech <- read_excel("/Users/sophieshan/Desktop/Group Project/1-Data/SWI_Master_Results.xlsx",
                     sheet = "Sheet1")
head(speech)
```

```
# A tibble: 6 x 29
  ID Cohort `Cleft Type` Sex Age L1 L2 L3 L_Group L1_SWI L2_SWI
  <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 10103     1         2     1 9.48 38 39 40         2 90.3 87.0
2 10106     1         2     1 9.01 38 39 40         2 89.0 85.1
3 10113     1         2     2 8.2 50 49 51         4 53.9 46.8
4 10116     1         2     1 6.89 48 52 53         5 86.4 85.1
5 10117     1         2     1 8.87 69 70 87        10 96.1 95.5
6 10121     1         2     1 7.65 54 55 56         6 84.4 91.6
# i 18 more variables: L3_SWI <dbl>, M_SWI <dbl>, L1_L2 <dbl>, L1_L3 <dbl>,
# L2_L3 <dbl>, L_Avg <dbl>, alv_pal <dbl>, alv_velar <dbl>, cl_sing <dbl>,
# con_null <dbl>, fr_bk <dbl>, fric_affr <dbl>, hi_lo <dbl>, r_w <dbl>,
# `st_fric-affr` <dbl>, st_nas <dbl>, vd_vl <dbl>, `VP Status` <dbl>
```

```
dim(speech)
```

```
[1] 115 29
```

Section 2: Prep Data for Analysis

Relabel Column Names

```
colnames(speech)[3] <- "Cleft_Type"
colnames(speech)[26] <- "st_fric_affr"
colnames(speech)[29] <- "VP_Status"
```

Make Variables into Factors

```

#rename study cohorts
speech$Cohort <- factor(
  speech$Cohort,
  levels = c(1, 2, 3, 4),
  labels = c("CLP", "CP", "TD", "OM")
)

#rename sex
speech$Sex <- factor(
  speech$Sex,
  levels = c(1, 2),
  labels = c("Male", "Female")
)

#rename VP status
speech$VP_Status <- factor(speech$VP_Status,
                           levels = c(1, 2),
                           labels = c("Adequate", "Inadequate"))

```

Create a Cohort x Sex Variable

```

speech$GroupSex <- factor(
  paste(speech$Cohort, speech$Sex),
  levels = c(
    "CLP Male", "CLP Female",
    "CP Male", "CP Female",
    "TD Male", "TD Female",
    "OM Male", "OM Female"
  )
)

```

Add Listener Group for ID 30310

```

#check to see which listeners are in for 30310
speech %>%
  filter(ID == "30310") %>%
  dplyr::select(L1, L2, L3)

```

```
# A tibble: 1 x 3
  L1    L2    L3
<dbl> <dbl> <dbl>
1    75    76    74
```

```
#check to see where listener 75, 76, 74 are
speech %>%
  filter(L1 == 75 | L1 == 76 | L1 == 74) %>%
  dplyr::select(ID, Cohort, Cleft_Type, Sex, Age, L1, L2, L3, L_Group)
```

```
# A tibble: 7 x 9
  ID Cohort Cleft_Type Sex    Age    L1    L2    L3 L_Group
<dbl> <fct>    <dbl> <fct>  <dbl> <dbl> <dbl> <dbl> <dbl>
1 10144 CP          5 Male   6.18   75   76   74    12
2 30130 CP          5 Male   6.48   74   83   76    12
3 20318 TD          1 Female 6.49   75   74   77    12
4 20319 TD          1 Female 7.06   76   74   78    12
5 30310 TD          1 Female 7.51   75   76   74    NA
6 20229 OM          1 Female 5.88   75   76   74    12
7 20230 OM          1 Male   6.41   75   76   74    12
```

```
#based on this, we see that the most likely listener group for ID 30310 is Group 12
speech[speech$ID == 30310, "L_Group"] <- 12
```

```
#make L_Group into a factor
speech <- speech %>%
  mutate(L_Group = as.factor(L_Group))
```

Create Long Version

```
#Extract variables for simple analysis
speech_subset <- speech %>%
  dplyr::select(ID, Cohort, Cleft_Type, Sex, Age,
                VP_Status, L1_SWI, L2_SWI, L3_SWI, L_Group)

#Transform Data to long format
speech_subset_long <- speech_subset %>%
  pivot_longer(
    cols = c(L1_SWI, L2_SWI, L3_SWI),
```

```

names_to = "Listener",

#Give unique names to each listener within group (there should be 66 unique listeners)
speech_subset_long <- speech_subset_long %>%
  mutate(Listener = paste0("G", L_Group, "_", str_extract(Listener, "L\\d+"))) %>%
  mutate(Listener = as.factor(Listener)) %>%
  mutate(L_Group = as.factor(L_Group))
dim(speech_subset_long)

```

```
[1] 345    9
```

```

#double check
n_distinct(speech_subset_long$Listener)

```

```
[1] 66
```

```

#Remove the row with one NA observation (L3 for ID 30310)
speech_subset_long <- speech_subset_long %>%
  filter(!(is.na(SWI == T)))
dim(speech_subset_long)

```

```
[1] 344    9
```

```

#summary
summary(speech_subset_long)

```

ID	Cohort	Cleft_Type	Sex	Age
Min. :10103	CLP: 69	Min. :1.000	Male :204	Min. :5.877
1st Qu.:10309	CP : 72	1st Qu.:1.000	Female:140	1st Qu.:6.605
Median :20219	TD :101	Median :1.000		Median :7.600
Mean :19566	OM :102	Mean :1.959		Mean :7.599
3rd Qu.:30107		3rd Qu.:2.000		3rd Qu.:8.449
Max. :30321		Max. :5.000		Max. :9.479

VP_Status	L_Group	Listener	SWI
Adequate :323	9 : 21	G12_L1 : 7	Min. : 40.26
Inadequate: 21	12 : 20	G12_L2 : 7	1st Qu.: 85.71
	1 : 18	G9_L1 : 7	Median : 90.26
	3 : 18	G9_L2 : 7	Mean : 87.85
	6 : 18	G9_L3 : 7	3rd Qu.: 93.51
	8 : 18	G1_L1 : 6	Max. :100.00
	(Other):231	(Other):303	

Section 3: Figures and Tables

Figure 1a

Distribution Speech of Mean Intelligibility Scores by Group and Sex Mean Speech Intelligibility vs. Age by Group. Boxplots show median, IQR, and range by group and sex.

```
# Custom x-axis positions: pairs are (Male, Female) for each group (CLP, CP, TD, OM)
# Gaps between pairs create visual separation between groups
positions4 <- c(1, 2.2, 4.8, 6.0, 8.6, 9.8, 12.4, 13.6)

# Draw boxplot with boxes placed at custom positions instead of default 1:8
# col alternates light/dark gray for Male/Female within each group
bp <- boxplot(M_SWI ~ GroupSex, data = speech,
              at = positions4,           # place boxes at custom x positions
              xaxt = "n",               # suppress default x-axis (added manually below)
              xlab = "",
              ylab = "Mean Speech Intelligibility Score",
              cex.main = 0.9,
              col = rep(c("gray85", "gray60"), 4), # light = Male, dark = Female
              medlwd = 3,                # thicker median line
              whisklty = 2,              # dashed whiskers
              boxwex = 0.9,              # box width
              xlim = c(0, 15)
)

# Add Male/Female labels below each box
text(
  x = positions4,
  y = par("usr")[3] - 3.5,              # slightly below the plot area
  labels = rep(c("Male", "Female"), 4),
  xpd = TRUE                            # allow drawing outside plot region
)

# Add group labels (CLP, CP, TD, OM) centered between each Male/Female pair
text(
  x = c(mean(c(1, 2.2)),
        mean(c(4.8, 6.0)),
        mean(c(8.6, 9.8)),
        mean(c(12.4, 13.6))),
  y = par("usr")[3] - 8,                 # further below than the Male/Female labels
  labels = c("CLP", "CP", "TD", "OM"),
```

```

xpd = TRUE
)

# Add x-axis title
mtext("Group and Sex", side = 1, line = 5)

# Overlay median values as text just above each median line (row 3 of bp$stats)
text(positions4, bp$stats[3, ] + 1,
      round(bp$stats[3, ], 1),
      cex = 0.7, font = 2)

```

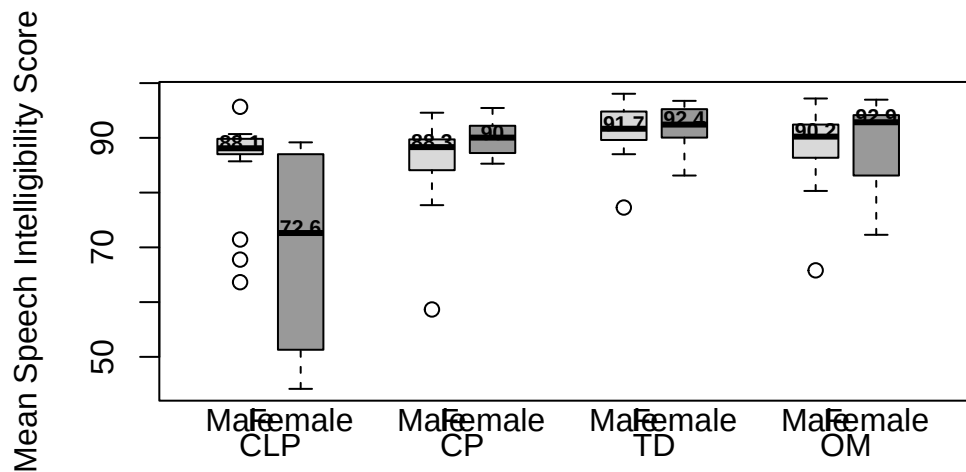


Figure 1b

Distribution Speech of Mean Intelligibility Scores by Group and Sex Mean Speech Intelligibility vs. Age by Group. Boxplots show median, IQR, and range by group and sex.

```

# Custom x-axis positions: pairs are (Male, Female) for each group (CLP, CP, Control)
# Gaps between pairs create visual separation between groups
positions3 <- c(1, 2.2, 4.8, 6.0, 8.6, 9.8)

# Merge TD and OM into a single Control group, then create a 3-level GroupSex variable
speech_3group <- speech %>%
  mutate(Cohort3 = case_when(
    Cohort == "CLP" ~ "CLP",
    Cohort == "CP" ~ "CP",
    Cohort %in% c("TD", "OM") ~ "Control"
  )) %>%

```

```

mutate(GroupSex3 = factor(
  paste(Cohort3, Sex),
  levels = c("CLP Male", "CLP Female",
             "CP Male", "CP Female",
             "Control Male", "Control Female")
))

# Draw boxplot with boxes placed at custom positions instead of default 1:6
# col alternates light/dark gray for Male/Female within each group
bp3 <- boxplot(M_SWI ~ GroupSex3, data = speech_3group,
               at = positions3,           # place boxes at custom x positions
               xaxt = "n",               # suppress default x-axis (added manually below)
               xlab = "",
               ylab = "Mean Speech Intelligibility Score",
               cex.main = 0.9,
               col = rep(c("gray85", "gray60"), 3), # light = Male, dark = Female
               medlwd = 3,               # thicker median line
               whisklty = 2,            # dashed whiskers
               boxwex = 0.9,            # box width
               xlim = c(0, 11)
)

# Add Male/Female labels below each box
text(
  x = positions3,
  y = par("usr")[3] - 3.5,             # slightly below the plot area
  labels = rep(c("Male", "Female"), 3),
  xpd = TRUE                          # allow drawing outside plot region
)

# Add group labels (CLP, CP, Control) centered between each Male/Female pair
text(
  x = c(mean(c(1, 2.2)),
        mean(c(4.8, 6.0)),
        mean(c(8.6, 9.8))),
  y = par("usr")[3] - 8,               # further below than the Male/Female labels
  labels = c("CLP", "CP", "Control"),
  xpd = TRUE
)

# Add x-axis title
mtext("Group and Sex", side = 1, line = 5)

```



```
# Overlay median values as text just above each median line (row 3 of bp3$stats)
text(positions3, bp3$stats[3, ] + 1,
      round(bp3$stats[3, ], 1),
      cex = 0.7, font = 2)
```

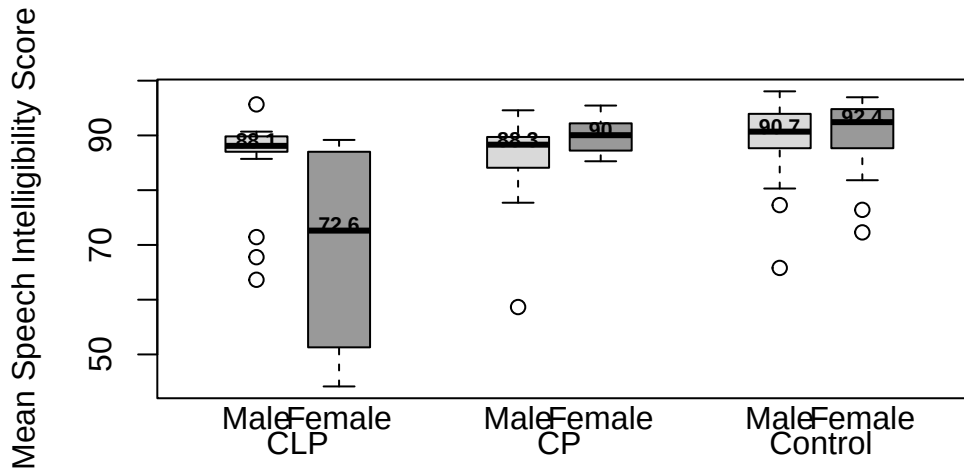


Figure 2

Scatter plot with points colored by group, points represent individuals. Optional trend lines may be added.

```
ggplot(speech, aes(x = Age, y = M_SWI, color = Cohort, shape = Cohort)) +
  geom_point(size = 2.5, alpha = 0.8) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    x = "Age",
    y = "Mean Speech Intelligibility Score",
    color = "Group",
    shape = "Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold")
  )
```

`geom\_smooth()` using formula = 'y ~ x'

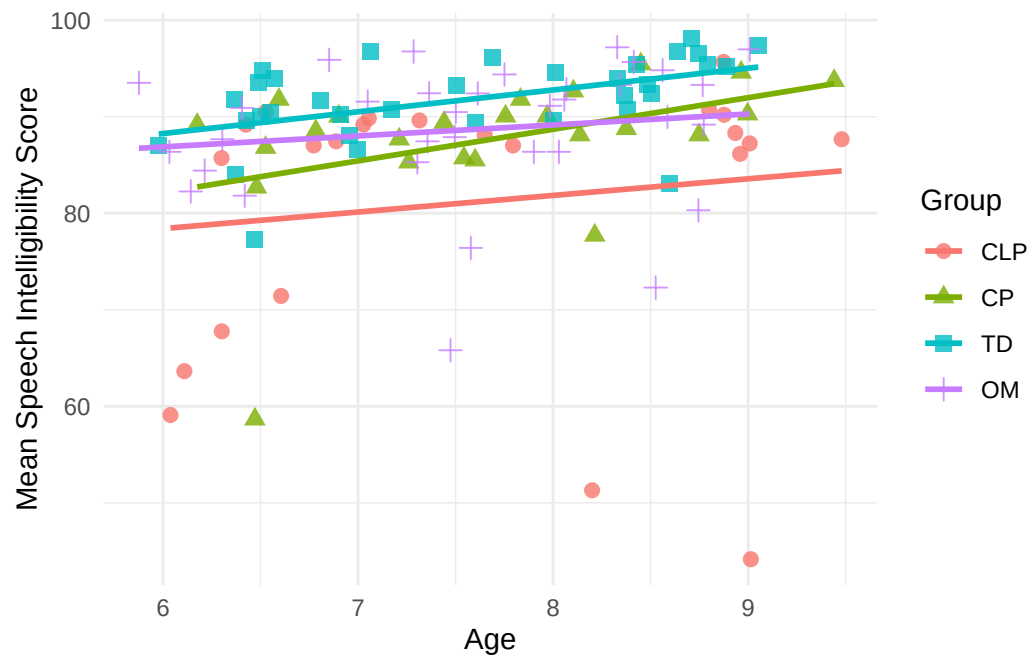
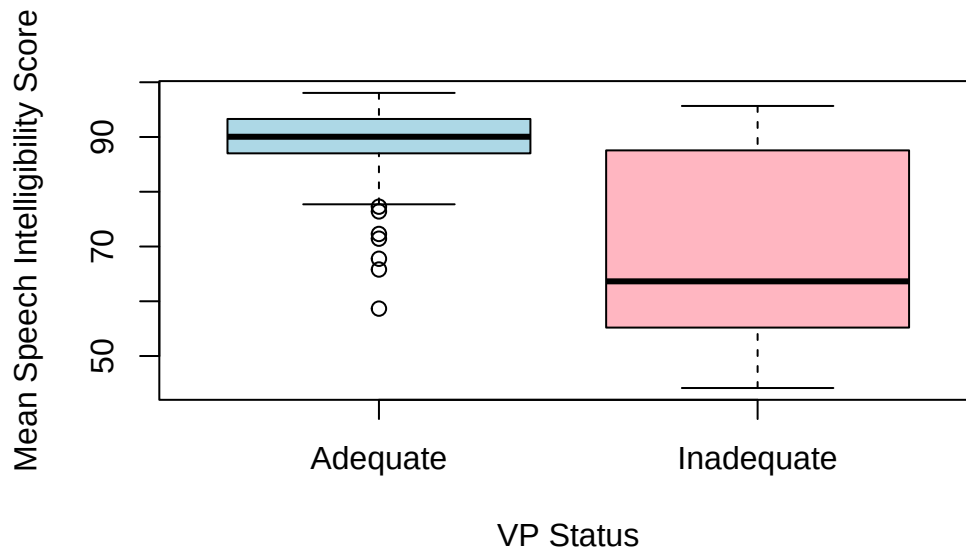


Figure 3

Boxplot split by mean intelligence scores by VP status.

```
boxplot(M_SWI ~ VP_Status,
  data = speech,
  xlab = "VP Status",
  ylab = "Mean Speech Intelligibility Score",
  col = c("lightblue", "lightpink"))
```



Section 4: Preliminary Analysis

ICC

To quantify observer agreement before running the main model. If the ICC is very high (> 0.90), averaging across observers and running a simpler ANOVA/linear regression okay.

Assume that for each listener, the same 3 listeners rated each of the study subjects. Note that in the raw data, groups 1,2,3,6,9,10,11,12,13,17,22 all had more than 3 listener IDs per listener group.

```
# Split data by listener group so ICC can be computed separately for each group
speech_L_Group <- speech %>%
  dplyr::select(L_Group, L1_SWI, L2_SWI, L3_SWI) %>%
  group_split(L_Group)

# Function to compute two-way agreement ICC for a single listener group
# Columns 2:4 are the three listener SWI scores; unit = "single" gives ICC for one rater
icc_per_group <- function(group_df){
  icc_val <- icc(group_df[, 2:4], model = "twoway",
                 type = "agreement", unit = "single")
  return(cbind(group_df$L_Group[1], icc_val$value))
}

# Apply function across all groups and combine into a single data frame
```

```
# V1 = group ID, V2 = ICC value
icc_results <- lapply(speech_L_Group, icc_per_group)
icc_results_df <- as.data.frame(do.call(rbind, icc_results))
icc_results_df$V2 <- round(icc_results_df$V2, 3)
icc_results_df
```

	V1	V2
1	1	0.773
2	2	0.234
3	3	0.443
4	4	0.983
5	5	0.811
6	6	0.744
7	7	0.075
8	8	0.964
9	9	0.861
10	10	0.806
11	11	0.973
12	12	0.547
13	13	0.717
14	14	0.975
15	15	0.930
16	16	0.602
17	17	0.963
18	18	0.572
19	19	0.947
20	20	0.691
21	21	0.761
22	22	0.959

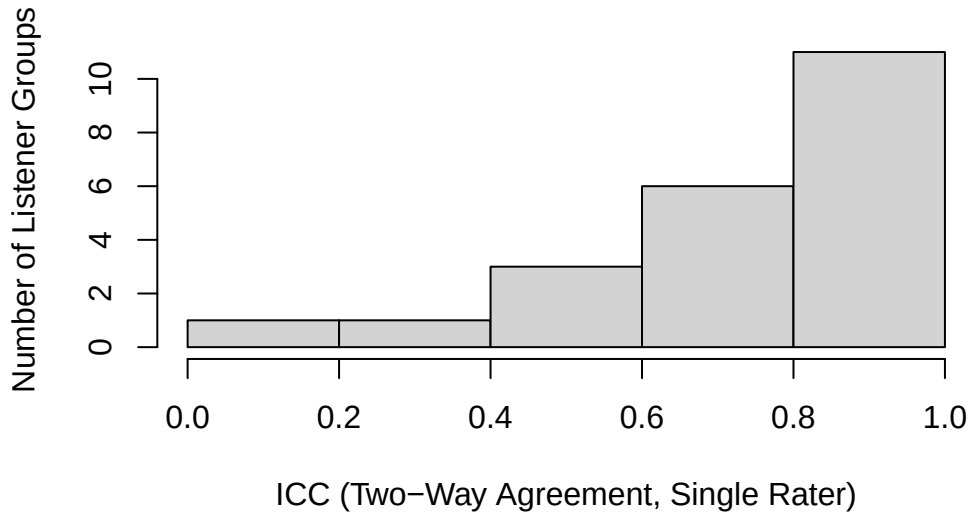
```
# Proportion of groups with poor agreement (ICC < 0.50)
prop.table(table(icc_results_df$V2 < 0.50))
```

	FALSE	TRUE
	0.8636364	0.1363636

```
# Distribution of ICC values across all listener groups
hist(icc_results_df$V2,
     main = "Distribution of ICC Values Across Listener Groups",
```

```
xlab = "ICC (Two-Way Agreement, Single Rater)",  
ylab = "Number of Listener Groups")
```

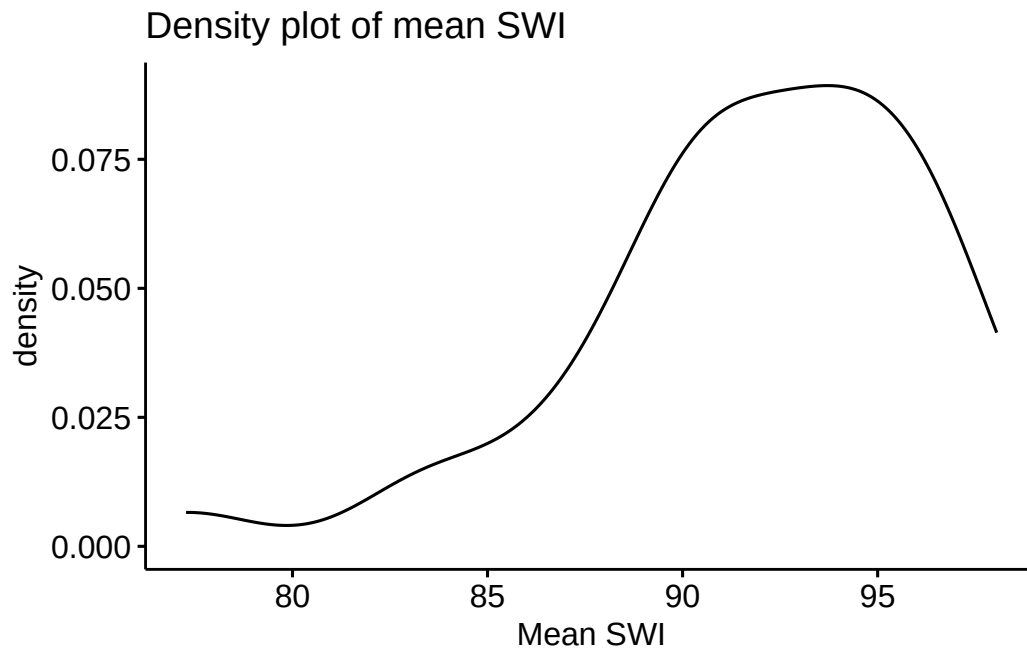
Distribution of ICC Values Across Listener Groups



Comparing OM to TD

A preliminary analysis was conducted to assess whether the TD and OM groups were sufficiently similar in mean SWI scores to justify pooling. Normality was first assessed using the Shapiro-Wilk test. According to Shapiro-Wilk normality test, the p-value > 0.05 implying that the distribution of the data are not significantly different from normal distribution. We cannot assume normality.

```
#TD  
TD <- speech %>%  
  filter(Cohort == "TD")  
ggdensity(TD$M_SWI,  
  main = "Density plot of mean SWI",  
  xlab = "Mean SWI")
```



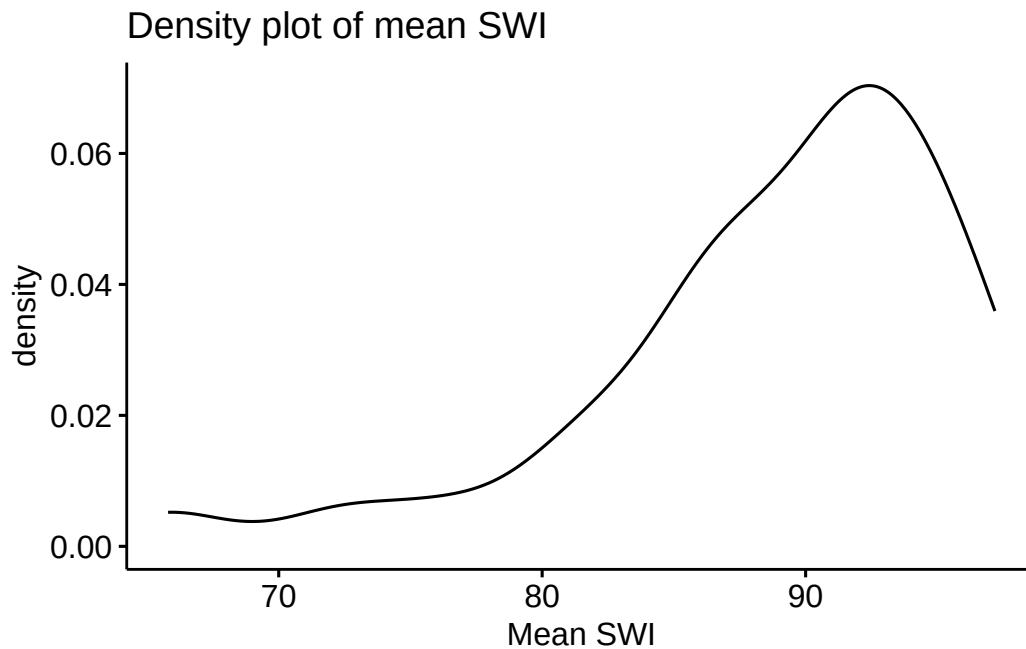
```
shapiro.test(TD$M_SWI)
```

Shapiro-Wilk normality test

data: TD\$M_SWI

W = 0.92041, p-value = 0.01659

```
#OM
OM <- speech %>%
  filter(Cohort == "OM")
ggdensity(OM$M_SWI,
  main = "Density plot of mean SWI",
  xlab = "Mean SWI")
```



```
shapiro.test(OM$M_SWI)
```

Shapiro-Wilk normality test

```
data: OM$M_SWI  
W = 0.88342, p-value = 0.001721
```

A bootstrapped equivalence test (TOST) was then conducted, as this approach allows us to actively test for similarity rather than simply failing to detect a difference. Bootstrapped methods were used given that normality assumptions were not met. An equivalence margin of ± 5 points was specified, representing a difference deemed negligible on the 0–100 SWI scale.

```
# Bootstrapped TOST: tests whether the mean difference falls within [-5, +5]  
# hypothesis = "EQU": equivalence hypothesis (both bounds must be satisfied)  
# paired = FALSE: independent samples (TD and OM are separate groups)  
# eqb = 5: equivalence margin of  $\pm 5$  points on the 0-100 SWI scale  
# alpha = 0.05: significance level for each one-sided test  
# boot_ci = "stud": studentized bootstrap confidence intervals  
# (more accurate than percentile CI)
```

```
set.seed(1)
```

```
boot_t_TOST(x = TD$M_SWI, y = OM$M_SWI,
            hypothesis = "EQU",
            paired = FALSE,
            eqb = 5,
            alpha = 0.05,
            boot_ci = c("stud"))
```

Bootstrapped Welch Two Sample t-test

The equivalence test was non-significant, $t(55.76) = -1.326$, $p = 1.01e-01$
 The null hypothesis test was significant, $t(55.76) = 2.135$, $p = 1.4e-02$
 NHST: reject null significance hypothesis that the effect is equal to zero
 TOST: don't reject null equivalence hypothesis

TOST Results

	t	df	p.value
t-test	2.135	55.76	0.014
TOST Lower	5.596	55.76	< 0.001
TOST Upper	-1.326	55.76	0.101

Effect Sizes

	Estimate	SE	C.I.	Conf. Level
Raw	3.0843	1.4023	[0.8695, 5.6845]	0.9
Hedges's g(av)	0.5109	0.2206	[0.1574, 0.883]	0.9

Note: studentized bootstrap ci method utilized.

The observed mean difference between TD and OM was 3.08 points (90% bootstrapped CI [0.87 5.68]). Formal equivalence within ± 5 points could not be confirmed as the upper bound was not satisfied ($p = .101$), however the difference was considered clinically negligible (~3% of the full scale range). The two groups were therefore combined into a single control group for all subsequent analyses.

```
#for wide dataset
speech_analysis <- speech %>%
  mutate(new_Cohort = if_else(Cohort == "TD" | Cohort == "OM",
                              "Control", Cohort))
table(speech_analysis$Cohort, speech_analysis$new_Cohort)
```

CLP Control CP

CLP	23	0	0
CP	0	0	24
TD	0	34	0
OM	0	34	0

```
#for long dataset
speech_long_analysis <- speech_subset_long %>%
  mutate(new_Cohort = if_else(Cohort == "TD" | Cohort == "OM",
                              "Control", Cohort))
table(speech_long_analysis$Cohort, speech_long_analysis$new_Cohort)
```

	CLP	Control	CP
CLP	69	0	0
CP	0	0	72
TD	0	101	0
OM	0	102	0

Examining the Influence of Inadequate VP Status

Now, we examine the influence of children with inadequate velopharyngeal (VP) status. We want to see if we should include them in the primary/secondary analysis (this was brought up as question by the investigator).

```
# Descriptive summary of mean SWI by VP status, cohort, and sex
# Used to quantify the disproportionate influence of
#inadequate VP children on CLP female estimates
speech_analysis %>%
  group_by(VP_Status, new_Cohort, Sex) %>%
  summarise(
    mean_SWI = round(mean(M_SWI), 1),
    sd       = round(sd(M_SWI), 2),
    n        = n(),
    .groups  = "drop"
  )
```

```
# A tibble: 9 x 6
  VP_Status new_Cohort Sex    mean_SWI    sd      n
  <fct>     <chr>     <fct>    <dbl> <dbl> <int>
1 Adequate CLP        Male      86   7.12   14
2 Adequate CLP        Female  87.4  1.56    3
```

3	Adequate	CP	Male	85.1	9.87	11
4	Adequate	CP	Female	89.8	3.18	12
5	Adequate	Control	Male	90	6.08	39
6	Adequate	Control	Female	90.5	6.25	29
7	Inadequate	CLP	Male	82.1	16.6	3
8	Inadequate	CLP	Female	51.5	7.47	3
9	Inadequate	CP	Male	88.1	NA	1

Children with inadequate VP were almost entirely concentrated in the CLP group and had substantially lower intelligibility scores than their adequate-VP peers. We see that the female with inadequate-VP status had nearly 36 points lower than adequate-VP CLP females. Inadequate-VP status and CLP is almost 1:1 (near-complete confounding). We decide to remove.

```
#for wide dataset
speech_analysis_no_vp <- speech_analysis %>%
  filter(VP_Status == "Adequate")
table(speech_analysis_no_vp$VP_Status)
```

```
Adequate Inadequate
      108          0
```

```
#for long dataset
speech_long_analysis_no_vp <- speech_long_analysis %>%
  filter(VP_Status == "Adequate")
table(speech_long_analysis_no_vp$VP_Status)
```

```
Adequate Inadequate
      323          0
```

Section 5: Aim 1

Table 1

```
table1(~ Cleft_Type + Sex + Age + VP_Status + M_SWI | new_Cohort,
       data = speech_analysis_no_vp)
```

	CLP	Control	CP	Overall
	(N=17)	(N=68)	(N=23)	(N=108)
Cleft_Type				
Mean (SD)	2.18 (0.393)	1.00 (0)	4.48 (0.511)	1.93 (1.43)
Median [Min, Max]	2.00 [2.00, 3.00]	1.00 [1.00, 1.00]	4.00 [4.00, 5.00]	1.00 [1.00, 5.00]
Sex				
Male	14 (82.4%)	39 (57.4%)	11 (47.8%)	64 (59.3%)
Female	3 (17.6%)	29 (42.6%)	12 (52.2%)	44 (40.7%)
Age				
Mean (SD)	7.64 (1.13)	7.57 (0.924)	7.65 (0.925)	7.60 (0.949)
Median [Min, Max]	7.32 [6.30, 9.48]	7.59 [5.88, 9.05]	7.60 [6.18, 9.44]	7.59 [5.88, 9.48]
VP_Status				
Adequate	17 (100%)	68 (100%)	23 (100%)	108 (100%)
Inadequate	0 (0%)	0 (0%)	0 (0%)	0 (0%)
M_SWI				
Mean (SD)	86.2 (6.47)	90.2 (6.11)	87.6 (7.43)	89.0 (6.60)
Median [Min, Max]	88.1 [67.8, 90.7]	91.6 [65.8, 98.1]	89.2 [58.7, 95.5]	90.0 [58.7, 98.1]

Primary Analysis

We use LRT-based forward selection (`add1()`) to determine which pairwise interactions to include in the primary model. None of the three candidate interactions were significant: `new_Cohort:Age` ($p=0.677$), `new_Cohort:Sex` ($p=0.517$), and `Age:Sex` ($p=0.398$). The final primary model retains main effects only.

```
# Step 1: Screen all pairwise interactions
# Fit main-effects-only base model with ML (REML cannot be used for LRT on fixed effects)
speech_lmm_base_ml <- lmer(SWI ~ new_Cohort + Age + Sex +
  (1 | ID),
  REML = FALSE,
  data = speech_long_analysis_no_vp)

# add1 adds each candidate interaction to the base model one at a time and tests
# improvement in fit via LRT - equivalent to forward selection for interactions
# Results: new_Cohort:Age (p=0.677), new_Cohort:Sex (p=0.517), Age:Sex (p=0.398)
# None are significant - retain main effects only
add1(speech_lmm_base_ml,
  scope = ~ (new_Cohort + Age + Sex)^2,
  test = "Chisq")
```

Single term additions

Model:

```
SWI ~ new_Cohort + Age + Sex + (1 | ID)
```

	Df	AIC	LRT	Pr(>Chi)
<none>		1861.1		
new_Cohort:Age	2	1864.3	0.78153	0.6765
new_Cohort:Sex	2	1863.8	1.31860	0.5172
Age:Sex	1	1862.4	0.71308	0.3984

Based on the LRT interaction screening above, the primary model does not need any interaction terms. Now we check which random effects structure is appropriate.

```
# Each child contributes 3 listener scores, so a child-level random intercept
# is needed to account for within-child correlation across listener ratings.
# Fit with both L_Group and ID to compare their variance contributions.
speech_lmm_re_check <- lmer(SWI ~ new_Cohort + Age + Sex +
                             (1 | L_Group) + (1 | ID),
                             REML = TRUE,
                             data = speech_long_analysis_no_vp)
```

boundary (singular) fit: see `help('isSingular')`

```
# ID variance (SD 5.96) dominates; L_Group variance (SD 0.60) is negligible
# once child-level clustering is accounted for - simplify to (1 | ID) only
VarCorr(speech_lmm_re_check)
```

Groups	Name	Std.Dev.
ID	(Intercept)	5.9689
L_Group	(Intercept)	0.0000
Residual		2.6831

The child-level random intercept (SD 5.96) accounts for ~82% of total variance, reflecting that each child has a strong personal baseline intelligibility. L_Group variance collapses to near zero once this is included, so the random effects structure is simplified to (1 | ID).

```
speech_lmm_final <- lmer(SWI ~ new_Cohort + Age + Sex +
                          (1 | ID),
                          REML = TRUE,
                          data = speech_long_analysis_no_vp)

summary(speech_lmm_final)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
lmerModLmerTest]

Formula: SWI ~ new_Cohort + Age + Sex + (1 | ID)

Data: speech_long_analysis_no_vp

REML criterion at convergence: 1837.8

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.8844	-0.5011	0.0458	0.4837	3.2655

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	35.628	5.969
Residual		7.199	2.683

Number of obs: 323, groups: ID, 108

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	70.1765	5.0310	102.9912	13.949	< 2e-16 ***
new_CohortControl	3.7983	1.7016	102.9971	2.232	0.02777 *
new_CohortCP	0.8805	2.0182	102.9945	0.436	0.66355
Age	2.0698	0.6292	102.9918	3.289	0.00137 **
SexFemale	1.3347	1.2391	103.0726	1.077	0.28392

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	nw_ChC	nw_CCP	Age
nw_ChrtCntr	-0.287			
new_CohrtCP	-0.221	0.690		
Age	-0.954	0.035	0.010	
SexFemale	0.005	-0.183	-0.212	-0.051

Let's compute the estimated marginal means (EMMs) for each cohort.

```
# EMMs are model-adjusted means that account for all other covariates (Age, Sex)

emm <- emmeans(speech_lmm_final, ~ new_Cohort)
emm
```

new_Cohort	emmean	SE	df	lower.CL	upper.CL
------------	--------	----	----	----------	----------

CLP	86.6	1.550	103	83.5	89.6
Control	90.4	0.754	103	88.9	91.9
CP	87.5	1.290	103	84.9	90.0

Results are averaged over the levels of: Sex
 Degrees-of-freedom method: kenward-roger
 Confidence level used: 0.95

Next, we look at the pairwise contrasts: Control vs CLP and CP vs CLP.

```
# ref = 1 sets CLP as the reference group (first level of new_Cohort)
# Hochberg adjustment controls familywise error rate across the two comparisons
emm_contrast <- contrast(emm, method = "trt.vs.ctrl", ref = 1, adjust = "hochberg")

# infer = TRUE returns estimates, 95% CIs, and p-values together
summary(emm_contrast, infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Control - CLP	3.798	1.70	103	-0.0723	7.67	2.232	0.0555
CP - CLP	0.881	2.02	103	-3.7103	5.47	0.436	0.6635

Results are averaged over the levels of: Sex
 Degrees-of-freedom method: kenward-roger
 Confidence level used: 0.95
 Conf-level adjustment: bonferroni method for 2 estimates
 P value adjustment: hochberg method for 2 tests

Now, we take a look at the linear mixed model diagnostics.

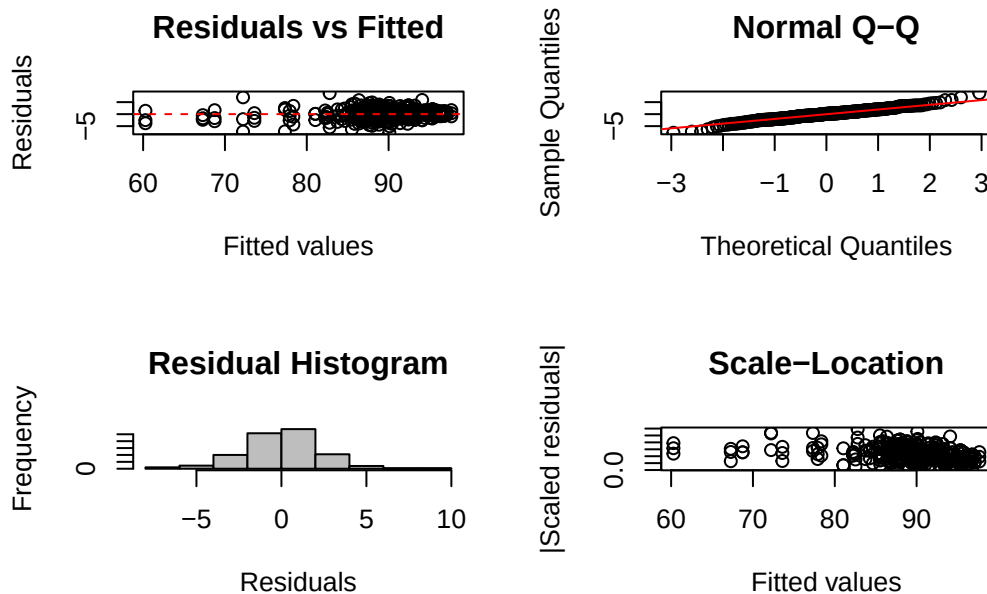
```
par(mfrow = c(2, 2))

# Residuals vs Fitted
plot(fitted(speech_lmm_final), residuals(speech_lmm_final),
     xlab = "Fitted values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, col = "red", lty = 2)

# Normal Q-Q
qqnorm(residuals(speech_lmm_final), main = "Normal Q-Q")
qqline(residuals(speech_lmm_final), col = "red")
```

```
# Residual Histogram
hist(residuals(speech_lmm_final),
     xlab = "Residuals", ylab = "Frequency",
     main = "Residual Histogram",
     col = "grey")

# Scale-Location
plot(fitted(speech_lmm_final), sqrt(abs(residuals(speech_lmm_final))),
     xlab = "Fitted values", ylab = "|Scaled residuals|",
     main = "Scale-Location")
```



```
par(mfrow = c(1, 1))
```

Sensitivity Analysis

Linear Regression with Mean SWI

This sensitivity analysis uses the child-level mean SWI (averaged across the 3 listeners) as the outcome instead of individual listener scores. Because each child contributes one observation, no random effect is needed. We fit the simplified model without L_Group and use an F-test to confirm it adds no explanatory value.

```
# Primary OLS model: cohort, age, and sex - mirrors the fixed-effects structure of the LMM
speech_ols_no_vp <- lm(M_SWI ~ new_Cohort + Age + Sex,
                      data = speech_analysis_no_vp)

summary(speech_ols_no_vp)
```

Call:

```
lm(formula = M_SWI ~ new_Cohort + Age + Sex, data = speech_analysis_no_vp)
```

Residuals:

Min	1Q	Median	3Q	Max
-25.793	-1.613	1.563	3.572	7.729

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	70.1770	5.0305	13.950	< 2e-16 ***
new_CohortControl	3.7991	1.7015	2.233	0.02772 *
new_CohortCP	0.8798	2.0180	0.436	0.66377
Age	2.0697	0.6292	3.290	0.00137 **
SexFemale	1.3368	1.2387	1.079	0.28304

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.166 on 103 degrees of freedom

Multiple R-squared: 0.1605, Adjusted R-squared: 0.1279

F-statistic: 4.922 on 4 and 103 DF, p-value: 0.001134

```
# Augmented model with L_Group as a fixed effect to test whether listener group
# explains additional variance in mean SWI beyond cohort, age, and sex
speech_ols_no_vp_lg <- lm(M_SWI ~ new_Cohort + Age + Sex + as.factor(L_Group),
                          data = speech_analysis_no_vp)
```

```
# F-test: L_Group adds no significant variance (p = 0.532) - consistent with
# the LMM finding that listener group variance collapses to near zero once
# child-level clustering is accounted for
anova(speech_ols_no_vp, speech_ols_no_vp_lg)
```

Analysis of Variance Table

Model 1: M_SWI ~ new_Cohort + Age + Sex


```
Model 2: M_SWI ~ new_Cohort + Age + Sex + as.factor(L_Group)
      Res.Df  RSS Df Sum of Sq    F Pr(>F)
1       103 3916.2
2        82 3150.2 21      765.93 0.9494 0.5319
```

```
# EMMs and pairwise contrasts from the simplified model, Hochberg-adjusted
emm_ols_no_vp <- emmeans(speech_ols_no_vp, ~ new_Cohort)

emm_ols_no_vp_contrast <- contrast(emm_ols_no_vp, method = "trt.vs.ctrl",
                                   ref = 1, adjust = "hochberg")
summary(emm_ols_no_vp_contrast, infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Control - CLP	3.80	1.70	103	-0.0711	7.67	2.233	0.0554
CP - CLP	0.88	2.02	103	-3.7105	5.47	0.436	0.6638

Results are averaged over the levels of: Sex
 Confidence level used: 0.95
 Conf-level adjustment: bonferroni method for 2 estimates
 P value adjustment: hochberg method for 2 tests

Including Inadequate VP Status

In the primary analysis, we excluded those 7 individuals with inadequate VP status. We now redo the analysis with them included. We include VP_Status as a main effect in the model but don't consider interaction effects with VP_Status because of singularity.

LMM

```
speech_lmm_vp <- lmer(SWI ~ new_Cohort + Age + Sex + VP_Status +
                      (1 | ID),
                      REML = TRUE,
                      data = speech_long_analysis)

summary(speech_lmm_vp)
```

Linear mixed model fit by REML. t-tests use Satterthwaite's method [
 lmerModLmerTest]
 Formula: SWI ~ new_Cohort + Age + Sex + VP_Status + (1 | ID)
 Data: speech_long_analysis

REML criterion at convergence: 2004.5

Scaled residuals:

Min	1Q	Median	3Q	Max
-2.8614	-0.4725	0.0484	0.4988	3.2181

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	55.63	7.459
	Residual	7.34	2.709

Number of obs: 344, groups: ID, 115

Fixed effects:

	Estimate	Std. Error	df	t value	Pr(> t)
(Intercept)	70.3550	5.9212	108.9906	11.882	< 2e-16 ***
new_CohortControl	5.1996	2.0519	108.9967	2.534	0.01270 *
new_CohortCP	3.1198	2.3724	108.9920	1.315	0.19127
Age	1.9918	0.7417	108.9912	2.686	0.00837 **
SexFemale	-0.9838	1.4715	109.0449	-0.669	0.50519
VP_StatusInadequate	-15.6495	3.2916	108.9908	-4.754	6.13e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	nw_ChC	nw_CCP	Age	SexFml
nw_ChrtCntr	-0.282				
new_CohrtCP	-0.207	0.671			
Age	-0.951	0.023	-0.007		
SexFemale	-0.003	-0.152	-0.171	-0.053	
VP_SttsIndq	-0.150	0.428	0.315	0.011	-0.078

We compute estimated marginal means.

```
# EMMs adjusted for Age, Sex, and VP_Status
emm_vp <- emmeans(speech_lmm_vp, ~ new_Cohort)
emm_vp
```

new_Cohort	emmean	SE	df	lower.CL	upper.CL
CLP	77.2	1.80	109	73.6	80.7
Control	82.4	1.88	109	78.6	86.1
CP	80.3	2.17	109	76.0	84.6

Results are averaged over the levels of: Sex, VP_Status
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95

Now, we compute pairwise contrasts using the same approach as the primary analysis, with Hochberg adjustment. The pairwise contrasts are extremely different from the primary analysis.

```
# Control vs CLP and CP vs CLP, Hochberg-adjusted
emm_vp_contrast <- contrast(emm_vp, method = "trt.vs.ctrl",
                             ref = 1, adjust = "hochberg")

# infer = TRUE returns estimates, 95% CIs, and p-values together
summary(emm_vp_contrast, infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Control - CLP	5.20	2.05	109	0.536	9.86	2.534	0.0254
CP - CLP	3.12	2.37	109	-2.272	8.51	1.315	0.1913

Results are averaged over the levels of: Sex, VP_Status
Degrees-of-freedom method: kenward-roger
Confidence level used: 0.95
Conf-level adjustment: bonferroni method for 2 estimates
P value adjustment: hochberg method for 2 tests

Now, let's take a look at the residuals.

```
par(mfrow = c(2, 2))

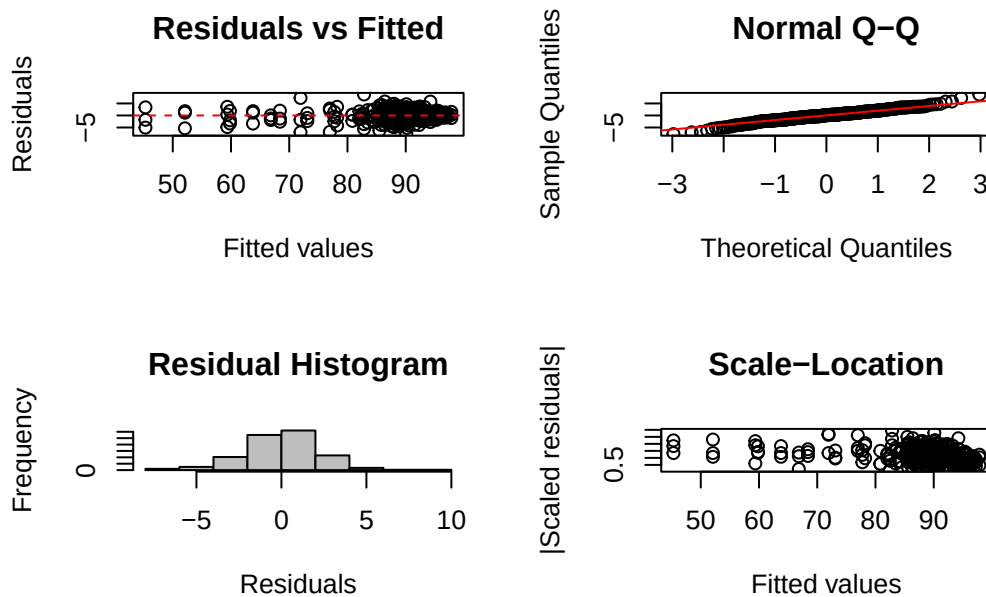
# Residuals vs Fitted
plot(fitted(speech_lmm_vp), residuals(speech_lmm_vp),
     xlab = "Fitted values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, col = "red", lty = 2)

# Normal Q-Q
qqnorm(residuals(speech_lmm_vp), main = "Normal Q-Q")
qqline(residuals(speech_lmm_vp), col = "red")

# Residual Histogram
```

```
hist(residuals(speech_lmm_vp),
     xlab = "Residuals", ylab = "Frequency",
     main = "Residual Histogram",
     col = "grey")

# Scale-Location
plot(fitted(speech_lmm_vp), sqrt(abs(residuals(speech_lmm_vp))),
     xlab = "Fitted values", ylab = "|Scaled residuals|",
     main = "Scale-Location")
```



```
par(mfrow = c(1, 1))
```

Linear Regression

We refit the OLS model on the full dataset with VP_Status included as a covariate, and again confirm that L_Group adds no explanatory value.

```
# OLS on child-level mean SWI with VP_Status as additional covariate
speech_ols <- lm(M_SWI ~ new_Cohort + Age + Sex + VP_Status,
                 data = speech_analysis)

summary(speech_ols)
```

Call:

```
lm(formula = M_SWI ~ new_Cohort + Age + Sex + VP_Status, data = speech_analysis)
```

Residuals:

Min	1Q	Median	3Q	Max
-27.705	-1.907	1.306	3.392	23.291

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	70.3554	5.9209	11.883	< 2e-16 ***
new_CohortControl	5.2005	2.0518	2.535	0.01268 *
new_CohortCP	3.1192	2.3723	1.315	0.19133
Age	1.9917	0.7416	2.686	0.00837 **
SexFemale	-0.9817	1.4712	-0.667	0.50599
VP_StatusInadequate	-15.6499	3.2915	-4.755	6.12e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.621 on 109 degrees of freedom

Multiple R-squared: 0.3362, Adjusted R-squared: 0.3057

F-statistic: 11.04 on 5 and 109 DF, p-value: 1.288e-08

```
# Augmented model with L_Group to verify it adds no variance in the full dataset
speech_ols_lg <- lm(M_SWI ~ new_Cohort + Age + Sex + VP_Status + as.factor(L_Group),
  data = speech_analysis)

# F-test: confirms L_Group is not needed even when inadequate VP children are included
anova(speech_ols, speech_ols_lg)
```

Analysis of Variance Table

Model 1: M_SWI ~ new_Cohort + Age + Sex + VP_Status

Model 2: M_SWI ~ new_Cohort + Age + Sex + VP_Status + as.factor(L_Group)

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	109	6329.9				
2	88	4909.3	21	1420.6	1.2126	0.2612

Now, we compute the marginal means and pairwise contrasts.

```
# EMMs and pairwise contrasts from the simplified model, Hochberg-adjusted
emm_ols <- emmeans(speech_ols, ~ new_Cohort)
```

```
emm_ols_contrast <- contrast(emm_ols, method = "trt.vs.ctrl", ref = 1,
                             adjust = "hochberg")
summary(emm_ols_contrast, infer = TRUE)
```

contrast	estimate	SE	df	lower.CL	upper.CL	t.ratio	p.value
Control - CLP	5.20	2.05	109	0.537	9.86	2.535	0.0254
CP - CLP	3.12	2.37	109	-2.273	8.51	1.315	0.1913

Results are averaged over the levels of: Sex, VP_Status

Confidence level used: 0.95

Conf-level adjustment: bonferroni method for 2 estimates

P value adjustment: hochberg method for 2 tests

Section 6: Aim 2

For Aim 2, we want to see if there are differences between the 11 phonetic contrasts for the three groups.

```
table1(~ alv_pal + alv_velar + cl_sing + con_null + fr_bk +
        fric_affr + hi_lo + r_w + st_fric_affr + st_nas + vd_vl
```

	CLP	Control	CP	Overall
	(N=17)	(N=68)	(N=23)	(N=108)
alv_pal				
Mean (SD)	82.0 (13.7)	90.8 (8.42)	86.9 (11.3)	88.6 (10.5)
Median [Min, Max]	85.7 [45.2, 95.2]	92.9 [61.9, 100]	88.1 [57.1, 100]	90.5 [45.2, 100]
alv_velar				
Mean (SD)	88.9 (9.06)	91.7 (7.82)	88.6 (9.32)	90.6 (8.40)
Median [Min, Max]	90.5 [61.9, 97.6]	92.9 [66.7, 100]	90.5 [57.1, 100]	92.9 [57.1, 100]
cl_sing				
Mean (SD)	88.2 (10.8)	93.4 (6.68)	89.1 (13.9)	91.7 (9.50)
Median [Min, Max]	90.5 [59.5, 100]	95.2 [76.2, 100]	92.9 [31.0, 100]	92.9 [31.0, 100]
con_null				
Mean (SD)	93.0 (6.57)	94.5 (5.24)	94.5 (7.20)	94.3 (5.88)
Median [Min, Max]	92.9 [76.2, 100]	95.2 [78.6, 100]	95.2 [66.7, 100]	95.2 [66.7, 100]
fr_bk				
Mean (SD)	81.6 (8.36)	84.4 (10.2)	79.1 (8.55)	82.8 (9.79)
Median [Min, Max]	83.3 [58.3, 92.9]	85.7 [54.8, 100]	79.4 [50.0, 90.5]	83.3 [50.0, 100]
fric_affr				
Mean (SD)	82.4 (10.5)	87.3 (10.5)	87.4 (9.90)	86.6 (10.4)
Median [Min, Max]	83.3 [61.9, 97.6]	90.5 [42.9, 100]	90.5 [61.9, 100]	90.5 [42.9, 100]
hi_lo				
Mean (SD)	84.2 (6.24)	88.9 (7.19)	86.0 (9.75)	87.5 (7.82)
Median [Min, Max]	83.3 [73.8, 92.9]	90.5 [51.3, 97.6]	90.5 [50.0, 95.2]	90.5 [50.0, 97.6]
r_w				
Mean (SD)	81.2 (17.1)	89.0 (13.6)	86.2 (15.7)	87.2 (14.8)
Median [Min, Max]	85.7 [33.3, 100]	92.9 [23.8, 100]	90.5 [40.5, 100]	90.5 [23.8, 100]
st_fric_affr				
Mean (SD)	86.7 (11.9)	90.4 (9.06)	85.0 (11.0)	88.7 (10.2)
Median [Min, Max]	90.5 [52.4, 97.6]	92.9 [66.7, 100]	88.1 [50.0, 97.6]	90.5 [50.0, 100]
st_nas				
Mean (SD)	88.8 (8.43)	93.3 (6.49)	92.2 (6.78)	92.4 (7.00)
Median [Min, Max]	90.5 [73.8, 100]	95.2 [74.4, 100]	95.2 [78.6, 100]	95.1 [73.8, 100]
vd_vl				
Mean (SD)	92.6 (5.18)	92.7 (8.27)	90.1 (10.9)	92.1 (8.51)
Median [Min, Max]	95.2 [78.6, 100]	95.2 [66.7, 100]	92.9 [47.6, 100]	95.2 [47.6, 100]

Primary Analysis

We define the 11 phonetic contrasts of interest and their corresponding human-readable labels used throughout the analysis.

```
# Column names in the dataset corresponding to each phonetic contrast
contrasts_vars <- c("alv_pal", "alv_velar", "cl_sing", "con_null",
  "fr_bk", "fric_affr", "hi_lo", "r_w",
  "st_fric_affr", "st_nas", "vd_vl")

# Human-readable labels for each contrast (used in output tables)
contrast_labels <- c("Alveolar-palatal sibilant (e.g., see-she)",
  "Alveolar-velar stop (e.g., tap-cap)",
  "Cluster-singleton (e.g., stop-top)",
  "Initial consonant-null (e.g., pat-at)",
  "Front-back vowel (e.g., fill-full)",
  "Fricative-affricate (e.g., ship-chip)",
  "High-low vowel (e.g., beat-bit)",
  "Liquid-glide (e.g., ray-way)",
  "Stop-fricative/affricate (e.g., base-vase)",
  "Stop-nasal (e.g., bop-mop)",
  "Voiced-voiceless stop (e.g., big-pig)")
```

A separate linear model is fit for each phonetic contrast, regressing the contrast score on cohort, sex, and age. Cohort EMMs are then extracted from each model and stacked into a single data frame for downstream testing.

```
# For each phonetic contrast outcome, fit a linear model with cohort, sex, and age
# No cohort:sex interaction - consistent with Aim 1 interaction screening
fit_contrast_model <- function(outcome_var, data) {
  formula <- as.formula(
    paste(outcome_var, "~ new_Cohort + Sex + Age")
  )
  lm(formula, data = data)
}

# Fit one model per contrast and name the list for downstream extraction
models <- map(contrasts_vars, fit_contrast_model, data = speech_analysis_no_vp)
names(models) <- contrasts_vars

# Extract cohort contrasts using emmeans, averaged over Sex
extract_coefs <- function(model, contrast_label) {
```



```

emm <- emmeans(model, ~ new_Cohort)
contrast(emm, method = "trt.vs.ctrl", ref = 1) %>%
  summary(infer = TRUE) %>%
  as_tibble() %>%
  mutate(contrast_var = contrast_label) %>%
  rename(conf.low = lower.CL, conf.high = upper.CL) %>%
  select(contrast_var, contrast, estimate, conf.low, conf.high, p.value)
}

# Stack contrasts from all models into one data frame
results_raw <- map2_dfr(models, contrast_labels, extract_coefs)

```

We apply a two-family multiple testing procedure with Hochberg correction. Family 1 tests CLP vs. Control for all 11 contrasts. Only contrasts significant in Family 1 are carried forward to Family 2, which tests CLP vs. CP. This sequential gating controls the familywise error rate across both comparisons.

```

# Family 1: CLP vs. Control across all 11 contrasts
# Hochberg correction applied across the 11 tests
family1 <- results_raw %>%
  filter(contrast == "Control - CLP") %>%
  arrange(p.value) %>%
  mutate(
    p_hochberg_f1 = p.adjust(p.value, method = "hochberg"),
    reject_f1      = p_hochberg_f1 < 0.05
  )

# Family 2: CLP vs. CP, only for contrasts significant in Family 1
rejected_contrasts <- family1 %>%
  filter(reject_f1) %>%
  select(contrast_var)

family2 <- results_raw %>%
  filter(contrast == "CP - CLP") %>%
  inner_join(rejected_contrasts, by = "contrast_var") %>%
  arrange(p.value) %>%
  mutate(
    p_hochberg_f2 = p.adjust(p.value, method = "hochberg"),
    reject_f2      = p_hochberg_f2 < 0.05
  )

```

Family 1 and Family 2 results are joined and formatted into a summary table displaying the

estimated group difference, 95% CI, and Hochberg-adjusted p-value for each phonetic contrast. Contrasts not carried into Family 2 are indicated with an em dash.

```
# Helper to format estimate and 95% CI as a single string
fmt_ci <- function(est, lo, hi) {
  sprintf("%.1f (%.1f, %.1f)", est, lo, hi)
}

# Join Family 1 and Family 2 results and format for display
table3 <- family1 %>%
  select(contrast_var,
         est_control = estimate,
         lo_control  = conf.low,
         hi_control  = conf.high,
         p_adj_f1    = p_hochberg_f1) %>%
  left_join(
    family2 %>%
      select(contrast_var,
             est_cp   = estimate,
             lo_cp    = conf.low,
             hi_cp    = conf.high,
             p_adj_f2 = p_hochberg_f2),
    by = "contrast_var"
  ) %>%
  arrange(match(contrast_var, contrast_labels)) %>%
  mutate(
    `CLP vs. Control: (95% CI)` = fmt_ci(est_control, lo_control, hi_control),
    `Adjusted p (Family 1)`     = ifelse(p_adj_f1 < 0.001,
                                         "<0.001",
                                         sprintf("%.3f", p_adj_f1)),
    `CLP vs. CP: (95% CI)`     = ifelse(
      !is.na(est_cp),
      fmt_ci(est_cp, lo_cp, hi_cp),
      "-"
    ),
    # Contrasts not carried into Family 2 are marked with an em dash
    `Adjusted p (Family 2)`     = case_when(
      is.na(p_adj_f2) ~ "-",
      p_adj_f2 < 0.001 ~ "<0.001",
      TRUE           ~ sprintf("%.3f", p_adj_f2)
    )
  ) %>%
  select(
```

```

`Phonetic Contrast`          = contrast_var,
`CLP vs. Control: (95% CI)`,
`Adjusted p (Family 1)`,
`CLP vs. CP: (95% CI)`,
`Adjusted p (Family 2)`
)

#display
table3 %>%
  kable(digits = 3, caption = "Phonetic Contrast Results")

```

Table 1: Phonetic Contrast Results

Phonetic Contrast	CLP vs. Control: (95% CI)	Adjusted p (Family 1)	CLP vs. CP: (95% CI)	Adjusted p (Family 2)
Alveolar-palatal sibilant (e.g., see-she)	8.3 (2.1, 14.5)	0.066	—	—
Alveolar-velar stop (e.g., tap-cap)	2.7 (-2.2, 7.7)	0.941	—	—
Cluster-singleton (e.g., stop-top)	4.3 (-1.1, 9.6)	0.923	—	—
Initial consonant-null (e.g., pat-at)	1.7 (-2.0, 5.3)	0.941	—	—
Front-back vowel (e.g., fill-full)	3.0 (-2.6, 8.5)	0.941	—	—
Fricative-affricate (e.g., ship-chip)	5.0 (-1.3, 11.4)	0.923	—	—
High-low vowel (e.g., beat-bit)	4.4 (-0.2, 9.0)	0.591	—	—
Liquid-glide (e.g., ray-way)	7.0 (-2.1, 16.2)	0.923	—	—
Stop-fricative/affricate (e.g., base-vase)	3.8 (-2.0, 9.6)	0.941	—	—
Stop-nasal (e.g., bop-mop)	4.1 (-0.0, 8.3)	0.511	—	—
Voiced-voiceless stop (e.g., big-pig)	-0.6 (-5.7, 4.5)	0.941	—	—